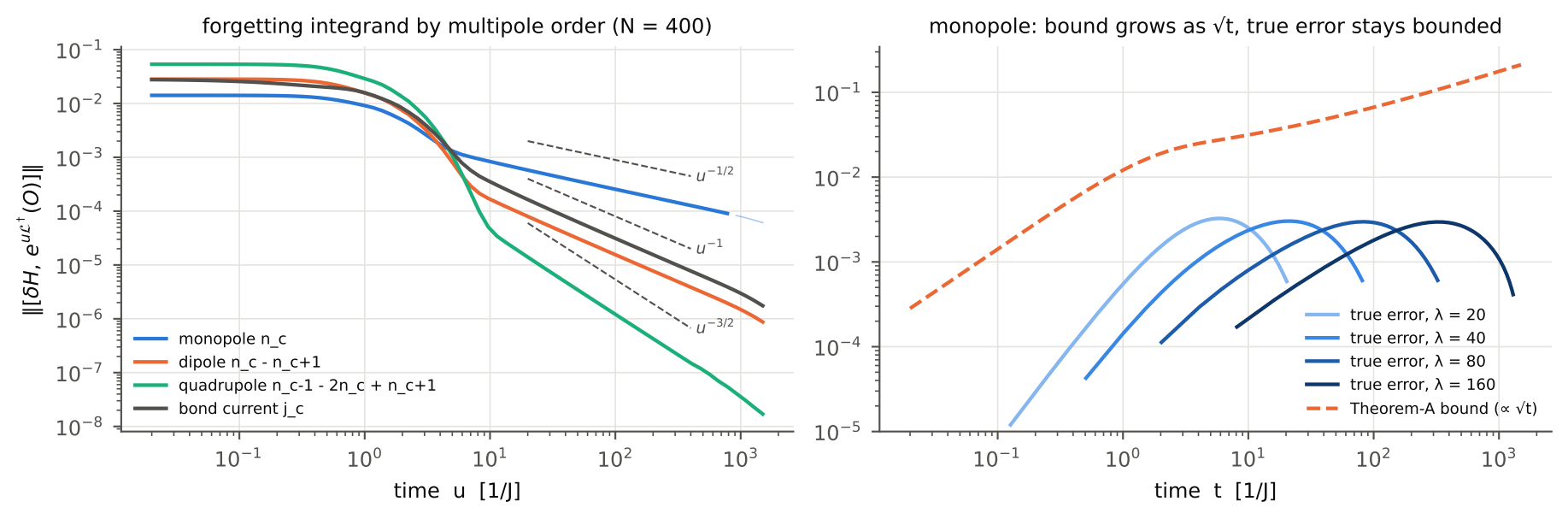
forgetting integrand by multipole order (N = 400)
$\|[\delta H,\ e^{u\mathcal{L}^\dagger}(O)]\|$
time u [1/J]
monopole n_c
dipole n_c - n_c+1
quadrupole n_c-1 - 2n_c + n_c+1
bond current j_c
$u^{-1/2}$
u^{-1}
$u^{-3/2}$
monopole: bound grows as √t, true error stays bounded
time t [1/J]
true error, λ = 20
true error, λ = 40
true error, λ = 80
true error, λ = 160
Theorem-A bound (∝ √t)